

GenAI Tool Evaluation: NotebookLM

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What the Tool Does

NotebookLM (Google) is a source-grounded research and study assistant. A user uploads their own material, such as notes, slides, PDFs, and links, and the tool answers questions, drafts summaries, and generates study aids using only that uploaded material as its sources, citing back to specific source passages. Its most distinctive feature is Audio Overview where two AI voices discuss, summarize, and make connections across the uploaded content. Unlike a general-purpose chatbot, its primary function is to synthesize and explain user-provided sources rather than generate independent content or perform computation.

SECTIONS-Based Analysis

Learners

The source-grounded design is a meaningful protection for agency compared with an open-ended chatbot, because the tool cannot wander into fabricated content unrelated to what the student or instructor actually provided. That protection is conditional, though, as NotebookLM does not evaluate the quality of the sources it is given. Agency is preserved to the extent that a student engages with the output (by studying and questioning it) rather than treating the generated content as a finished product.

Teaching Functions

The tool meaningfully supports review and consolidation of material that a student has already engaged with in class. In particular, it supplies a second voice to describe the content, which can be useful before an exam or a supplement to reading. It does however risk shortcutting the learning process if it's used to generate a solution to a specific problem rather than to build understanding of a concept.

Interaction

The Audio Overview feature of NotebookLM is largely one-directional, where the student listens rather than participates in real time. While there is a voice participation feature currently in beta testing, it doesn't reliably replicate the back-and-forth of office hours or a study group, and it could reduce a student's inclination to seek out those higher-value human interactions if it's treated as a substitute rather than a supplement. The teacher does, after all, have knowledge that extends beyond the notes and readings that are supplied in class.

Organizational Issues & Security

Although it has its limits, there is a free tier available, and an institutional deployment exists through Google Workspace for Education. This deployment carries documented assurances that data is not used for model training and is not human-reviewed under education accounts. Practical limits with the tool are well known and each implementation should be tested before use in the classroom. In particular, uploaded PDFs that contain embedded images of math notation can render incorrectly when the tool tries to read the document.

Learning-Theory Analysis

Psychoeducational Principles (Salinas, 2008)

As impressive as it is, when Audio Overview is used only for passive listening, the tool is most similar to the knowledge and comprehension level of the CL/IC model. While this isn't necessarily problematic, leaving the tool at this level wastes its potential. For instance, when follow-up tasks are implemented, such as asking students to explain back the concept or apply the knowledge to a new example, this tool comfortably sits in the application and analysis level.

Meaning-Making and Transfer (Holmes, Bialik, & Fadel, 2019)

Mathematics is often taught as a sequence of steps in order to obtain the correct answer. The sort of knowledge learned in this case would be characterized by Holmes, et. al. as fragile and nontransferable. Mathematics is more accurately understood as a framework of thinking or a structure for understanding, and a powerful way of teaching this is to present mathematics almost like a story. Do so creates relational knowledge that is both durable and transferable.

NotebookLM can support this kind of learning when it is used with (teacher provided) source material that already emphasizes conceptual relationships. It's capable of acting like a second-voice or a tutor that re-describes the relationships they've already learned. This added exposure can strengthen the familiarity and understanding of the topic.

Risks and Design Safeguards

NotebookLM can make passive review feel more complete than it actually is. A polished summary or Audio Overview may create familiarity with a topic without raising the student through Bloom's taxonomy. This risk is greater when the source material is procedural or when students use the tool to obtain solutions rather than to review conceptual relationships. Instructor-provided materials should therefore emphasize concepts, connections, and reasoning rather than isolated procedures, and students should be discouraged from using the tool as a substitute for working through problems themselves. A brief production task after engaging with the tool provides an additional safeguard by requiring students to move from receiving information to working with it.

Educational Use Cases

- Structured pre-class or pre-exam review of instructor-curated unit notes, especially when the Audio Overview or Explainer Video feature paired with a short task that engages students more deeply with the content.
- Review for students who benefit from repetition, auditory presentation, or captioned content.
- Re-entry into course material after an absence or before a cumulative assessment..

Recommendation — Use with Limits: NotebookLM is a strong option for structured review when it is used with instructor-curated, conceptually focused material and paired with an activity that requires students to explain, apply, or extend what they reviewed. Its value is much weaker when used for passive listening, procedural problem solving, or unsupervised study from unvetted sources. I would recommend it as a supplemental review tool rather than as a replacement for active practice, discussion, or independent reasoning.